

Coursework details

1. Input Modalities

In this part, you will implement systems using **Python** that apply:

- **Image processing**
- **Sound processing**
- **Text analysis** (Natural Language Processing – NLP)

Each modality involves handling **different types of data** and requires specific processing techniques. These modalities should be applied within a context of your choice. For example:

- Identifying emotions
- Analysing animal interactions
- Detecting diseases
- Identifying personality traits

You are free to select the application domain, provided that **all three modalities are meaningfully integrated** to achieve a cohesive system.

2. Study

it aims to determine whether the **adaptive system** enhances **user experience, engagement, and learning performance** compared to a conventional non-adaptive system.

Data Collection and System Comparison

You will collect data from **two educational systems designed to teach Python**:

1. **Adaptive educational system based on agentic AI**
2. **Static question-and-answer (non-adaptive) system**

These systems will be compared in terms of:

- **Usability**
- **Cognitive workload**
- **Learning performance**
- **User engagement**
- **Trust and explainability**

Participants

- Collect data from **16 students in Year 2 and Year 3**
- Participants will be split into two groups:
 - **8 students** use the static (simple) system
 - **8 students** use the adaptive (agentic AI) system

Procedure

- Both groups will answer the **same set of Python-related questions**
- The systems will automatically log:
 - Eye-gaze data
 - Facial emotion data
 - Time taken to answer each question
 - Whether answers are correct or incorrect

Retrospective Evaluation

After interacting with the assigned system, participants will complete a retrospective evaluation that includes:

- **Descriptive comments:** Participants can provide qualitative feedback on their experience, suggestions, or any difficulties they encountered.
- **System Usability Scale (SUS):** To measure perceived usability of the system (**on e-learning questions and excel sheet**).
- **NASA-TLX:** To assess cognitive workload during the **interaction (on e-learning questions and excel sheet)**.
- **Trust and Explainability Questionnaire:** To evaluate participants' trust in the system and their understanding of the AI's guidance or feedback.

#	Question	1	2	3	4	5	6	7	8	9	10
1	I trust the AI to provide helpful guidance on Python problems.	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
2	The AI seems knowledgeable and reliable in its suggestions.	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
3	I understand why the AI provided this hint or feedback.	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
4	The AI's explanation helped me understand my mistake.	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

Scale: 1 = Strongly Disagree, 10 = Strongly Agree

3. Data Analysis and Visualisation

You will analyse the collected data using appropriate methods and present the results using **clear and informative visualisations**.

Measure	What to Calculate	Suggested Visualization
SUS (System Usability Scale)	Mean and standard deviation of SUS scores	Bar chart or box plot comparing groups
NASA-TLX (Cognitive Workload)	Mean and standard deviation of overall workload	Bar chart or box plot comparing groups
Trust & Explainability Questionnaire	Mean and standard deviation for each question (1–10 scale)	Bar chart for each question comparing groups

4. Report (10%)

The report should include the following:

1. **Multimodality**
 - Describe the different modalities used (image, sound, text)
 - Explain their purpose in the system
 - Summarize the results obtained from applying these modalities
2. **Study and Data Analysis**
 - Explain the study design and data collection process
 - Present the analysis of the collected data
3. **Discussion of Results**
 - Comment on the study findings
 - Reflect on insights, patterns, or differences observed between the systems

Submissions

First submission (Week 9) 40%

- **Report with this part (Multimodality)**
 2. Describe the different modalities used (image, sound, text)
 3. Explain the purpose of the models in the system
 4. Summarize the results obtained from applying these modalities

2. Data used

3. Code

Second Submission (Week 11)

- **Complete report (PDF)**

The report should include the following:

1. Multimodality
 - Describe the different modalities used (image, sound, text)
 - Explain their purpose in the system
 - Summarize the results obtained from applying these modalities
 2. Study and Data Analysis
 - Explain the study design and data collection process
 - Present the analysis of the collected data
 3. Discussion of Results
 - Comment on the study findings
 - Reflect on insights, patterns, or differences observed between the systems
- **Raw Data of the**
 1. **logged files for each participant for each system static and adaptive** (Excel sheets)
 2. **System Usability Scale (SUS): answers of each for each system static and adaptive** (Excel sheets)
 3. **NASA-TLX: answers of each for each system static and adaptive** (Excel sheets)
 4. **Trust and Explainability Questionnaire: answers of each for each system static and adaptive** (Excel sheets)